

A COUNTEREXAMPLE TO TARANENKO'S DIAGONAL-DECOMPOSITION CONJECTURE

ABSTRACT. Taranenko conjectured that every polystochastic matrix of even dimension is a nonnegative linear combination of diagonal indicators. We show that the symmetric vertex $A_4 \in \Omega_3^4$ from Taranenko's 2025 enumeration is not in this cone. The proof is an explicit linear functional that is positive on A_4 and nonpositive on every diagonal indicator. This disproves Taranenko's Conjecture 4, restated by Child and Wanless as Conjecture 1.9. The pair $(d, n) = (4, 3)$ is componentwise minimal.

Let $I_n = \{0, \dots, n-1\}$. A d -dimensional matrix of order n is an array $X = (X_x)_{x \in I_n^d}$. It is *polystochastic* if $X_x \geq 0$ and every line has sum 1; write Ω_n^d for the set of such arrays. A *diagonal* is a set $D \subset I_n^d$ of size n such that every coordinate projection maps D bijectively onto I_n . Let P_D be its indicator array and set

$$\mathcal{C}_n^d = \text{cone}\{P_D : D \text{ is a diagonal in } I_n^d\}.$$

The arrays P_D are the $(d-1)$ -permutation matrices in the terminology of Child and Wanless [1].

Taranenko conjectured that $\Omega_n^d \subseteq \mathcal{C}_n^d$ whenever d is even [2, Conjecture 4]. Child and Wanless restated this as Conjecture 1.9 in [1]; it also appears as Conjecture 16 in Taranenko's 2025 dissertation [3]. We give a counterexample in Ω_3^4 .

Put $I = I_3$. For $x \in I^4$, define

$$\nu(x) = (\nu_0(x), \nu_1(x), \nu_2(x)), \quad \nu_r(x) = |\{j : x_j = r\}|.$$

Let $A = (A_x)_{x \in I^4}$ be given by

$$(1) \quad A_x = \begin{cases} 1, & \nu(x) \in \{(4, 0, 0), (0, 4, 0)\}, \\ \frac{1}{2}, & \nu(x) \in \mathcal{H}, \\ 0, & \text{otherwise,} \end{cases}$$

where

$$(2) \quad \mathcal{H} = \{(0, 1, 3), (0, 2, 2), (1, 0, 3), (1, 2, 1), (2, 0, 2), (2, 1, 1), (2, 2, 0)\}.$$

With rows indexed lexicographically by (x_1, x_2) and columns by (x_3, x_4) , this is the symmetric vertex A_4 in [4, Appendix, item 4].

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Theorem 1. *The array A belongs to $\Omega_3^4 \setminus \mathcal{C}_3^4$. Consequently, Conjecture 4 of [2] and Conjecture 1.9 of [1] are false.*

Proof. Fix three coordinates and let $\mu = (\mu_0, \mu_1, \mu_2)$ record their multiplicities. If e_r is the r th standard basis vector, then the three cells in the resulting line have types $\mu + e_0$, $\mu + e_1$, and $\mu + e_2$. For $\mu = (3, 0, 0)$ and $(0, 3, 0)$, the corresponding values are respectively $(1, 0, 0)$ and $(0, 1, 0)$. For each of the other eight weak compositions of 3, they are a permutation of $(\frac{1}{2}, \frac{1}{2}, 0)$. Thus every line has sum 1, and $A \in \Omega_3^4$.

Let $S = \text{supp}(A)$ and define

$$S_+ = \{x : \nu(x) \in \{(0, 1, 3), (1, 0, 3)\}\}, \quad S_- = \{x : \nu(x) = (2, 2, 0)\}.$$

For any array X , set

$$(3) \quad g(X) = \sum_{x \in S_+} X_x - \sum_{x \in S_-} X_x - \sum_{x \notin S} X_x.$$

Since $|S_+| = 2\binom{4}{1} = 8$ and $|S_-| = \binom{4}{2} = 6$, all at value $\frac{1}{2}$ in A ,

$$(4) \quad g(A) = 8 \cdot \frac{1}{2} - 6 \cdot \frac{1}{2} = 1.$$

We show that $g(P_D) \leq 0$ for every diagonal D . Every cell in S_+ has three coordinates equal to 2. Hence any two cells in S_+ agree in at least one coordinate, so a diagonal contains at most one of them. It follows immediately that $g(P_D) \leq 0$ if $D \cap S_+ = \emptyset$. The same holds if $D \cap S_+ \neq \emptyset$ but $D \not\subseteq S$: the possible contribution $+1$ from S_+ is offset by at least one contribution -1 from $I^4 \setminus S$.

It remains to consider a diagonal $D \subseteq S$ meeting S_+ . The supported types are

$$\mathcal{T} = \{(4, 0, 0), (0, 4, 0), (0, 1, 3), (0, 2, 2), (1, 0, 3), \\ (1, 2, 1), (2, 0, 2), (2, 1, 1), (2, 2, 0)\}.$$

The three type vectors of the cells of D sum to $(4, 4, 4)$, because each coordinate position contains each symbol once along D . After fixing the S_+ -cell, the ν_2 -components of the other two types sum to 1. Among the supported types, those with $\nu_2 = 0$ are

$$(4, 0, 0), \quad (0, 4, 0), \quad (2, 2, 0),$$

and those with $\nu_2 = 1$ are

$$(1, 2, 1), \quad (2, 1, 1).$$

Matching the ν_0 - and ν_1 -components shows that the remaining types are

$$\begin{cases} (2, 1, 1) \text{ and } (2, 2, 0), & \text{if the } S_+\text{-type is } (0, 1, 3), \\ (1, 2, 1) \text{ and } (2, 2, 0), & \text{if the } S_+\text{-type is } (1, 0, 3). \end{cases}$$

Thus every supported diagonal meeting S_+ also meets S_- , and again $g(P_D) \leq 0$.

If $A = \sum_D c_D P_D$ with $c_D \geq 0$, then linearity and (4) give

$$1 = g(A) = \sum_D c_D g(P_D) \leq 0,$$

a contradiction. $\square$

Proposition 2. *Within the class of even-dimensional polystochastic matrices, no counterexample exists when $d = 2$ or $n \leq 2$. Hence $(d, n) = (4, 3)$ is componentwise minimal.*

Proof. For $d = 2$, the assertion is the Birkhoff–von Neumann theorem. Order 1 is trivial. Now let X have order 2 and even dimension d . Index its cells by $x \in \{0, 1\}^d$ and put $t = X_{(0, \dots, 0)} \in [0, 1]$. Since the two entries in every line sum to 1, flipping one coordinate replaces a cell value by its complement. Hence

$$X_x = \begin{cases} t, & \sum_i x_i \equiv 0 \pmod{2}, \\ 1 - t, & \sum_i x_i \equiv 1 \pmod{2}. \end{cases}$$

For $\bar{x} = (1 - x_1, \dots, 1 - x_d)$, the pair $\{x, \bar{x}\}$ is a diagonal. Because d is even, x and $\bar{x}$ have the same parity. The complementary pairs partition each parity class, and therefore

$$X = t \sum_{\{x, \bar{x}\} \text{ of even parity}} P_{\{x, \bar{x}\}} + (1 - t) \sum_{\{x, \bar{x}\} \text{ of odd parity}} P_{\{x, \bar{x}\}},$$

where each unordered pair is counted once. $\square$

Remark 3 (Scope and provenance). The conjecture imposes no parity condition on the order, so the odd order of A is permitted. This example does not address the additional variant in which the order is also required to be even. The array A itself is not new; the contribution here is the separating certificate (3). Taranenko records $\text{per}(A) = 33/4 > 0$ [4, Appendix, item 4], so the example does not concern the separate positive-permanent conjecture.

REFERENCES

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